Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations

Anonymous Author(s)

Abstract

Quoted stablecoin price gaps are not trades. We introduce **Stablecoin StressBench**, a benchmark showing that most apparent stablecoin dislocations are optical rather than executable: during March 2023, 34.3% of minutes show a $>$ 10 bps USDC basis, but only 2.88% yield net profit after execution costs, a **12× gap** (block bootstrap 95 % CI: 8×–24×). No model trained on calm-market data closes this gap; the hindsight oracle gap exceeds 260 bps. Cross-mechanism meta-labeling trained on Terra/LUNA achieves +82.5 bps on the SVB test split (50.8% of oracle returns), providing evidence that order-book microstructure is transferable across structurally distinct stress mechanisms. A three-way RL diagnostic distinguishes density from supervision: the unconditioned PPO-GRU (2.3% positive rate) degenerates to NoTrade; a conditioned PPO-GRU (17% rate, primary-signal windows only) trades 919 times but earns −29.2 bps; only meta-labeling's explicit positive-label supervision reaches +82.5 bps. To the best of our knowledge, Stablecoin StressBench is the first public benchmark for stablecoin stress events that labels one-minute dislocations using executable route simulation, combining documented route-level depth provenance, VWAP book-walking, explicit fee and settlement penalties, a hindsight executable-profit ceiling, and cross-mechanism transfer evaluation, rather than quoted price gaps alone.

Keywords

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

ACM Reference Format:

Anonymous Author(s). 2026. Stablecoin StressBench: An Execution-Aware Benchmark for Stablecoin Dislocations. In *Proceedings of ACM International Conference on AI in Finance (ICAIF '26).* ACM, New York, NY, USA, **??** pages.

1 Introduction

Stablecoins are dollar-pegged cryptocurrencies that underpin the majority of DeFi lending, trading, and payments. During stress events, peg fragmentation creates cross-venue price gaps that look like arbitrage opportunities: buy cheaply on one exchange, sell at par on another, capture the spread. In practice, however, a quoted price gap is not a trade. Clearing the required order-book depth on a large notional moves the execution price; VWAP slippage, taker fees, and CEX-to-CEX settlement latency each add frictions that are small in isolation but jointly eliminate most of the apparent profit. The result is that many stablecoin dislocations are illusory: they appear profitable on a price chart but disappear once the cost of executing a real order through a live order book is accounted for. We call these *optical* dislocations, to distinguish them from *executable* ones where net profit survives all transaction costs.

Prior machine-learning work on stablecoin predictability uses price-based labels to define "profitable" windows, which conflates the optical and executable layers and systematically overstates how predictable actual trading opportunities are. **Stablecoin StressBench** is designed to close this gap. It provides per-minute execution-aware labels that separate three dislocation layers: (1) an *optical* dislocation (quoted basis exceeds a threshold), (2) an *executable* dislocation (net profit is positive after VWAP execution, fees, and settlement at a specified notional), and (3) a *predictable* dislocation (a model correctly identifies the window ex ante). The distance between these layers is the paper's core measurement.

We build the benchmark on the March 2023 USDC/SVB stress event, where Silicon Valley Bank's collapse triggered a brief USDC depeg. Using Binance L2 order-book data[1] and a full VWAP book-walking model, we find that 34.3% of 1-minute windows show a $>$ 10 bps price basis but only 2.88% remain net-profitable at \$10K notional: a 12× price-to-execution gap. We then construct a model ladder spanning rule-based thresholds, statistical baselines, gradient-boosted classifiers, sequence models (GRU), and meta-labeling filters, evaluating each model's ability to identify the 2.88% of executable windows ex ante. Every model trained on calm-control data loses money on executable arbitrage tasks; the hindsight oracle gap exceeds 260 bps. Cross-mechanism meta-labeling, trained on a different stress event (Terra/LUNA 2022), is the only approach that achieves positive net returns (+82.5 bps, 50.8% oracle capture).

Contributions.

1. **Execution-aware benchmark:** per-minute labels with documented order-book depth provenance, VWAP book-walking, fees, notional scaling, and a hindsight oracle ceiling. (§**??**; Table **??**)
2. **18-event stress taxonomy:** seven mechanism classes (2020–2023) with data-tier governance of permitted claims. Terra/LUNA cross-mechanism check gives 5.9× vs. the Tier-A 12× ratio. (§**??**, Table **??**; §**??**, Table **??**)
3. **Oracle gap evidence:** +162 bps oracle vs. +17.2 bps best model (basis task), $>$ 260 bps gap on executable arbitrage (structural, not label-mismatch), decomposed into FP cost, timing, and sizing components. (§**??**, Table **??**; §**??**)
4. **Cross-mechanism transfer and RL diagnostics:** Meta-labeling trained on Terra/LUNA achieves +82.5 bps (50.8% oracle capture) on SVB. Conditioned and unconditioned PPO-GRU experiments establish that density is necessary but explicit positive-label supervision is additionally required: conditioned PPO-GRU at 17% density earns −29.2 bps vs. meta-labeling's +82.5 bps. (§**??**, Table **??**; §**??**, Table **??**)

ICAIF '26, Milan, Italy
2026.

[1] The BTC-USDC buy leg uses kline-proxy depth for the SVB window; all other legs use real L2. The 20% depth haircut sensitivity in §**??** bounds this approximation.

Paper organisation. Section ?? reviews related work. Section ?? describes benchmark design and data. Section ?? defines execution-aware labels. Section ?? describes the model ladder and evaluation. Section ?? presents all results, including the oracle gap, cross-mechanism transfer, SHAP analysis, oracle gap decomposition, and RL execution policy evaluation. Section ?? discusses limitations. Section ?? concludes.

2 Related Work

Stablecoin stress. Uhlig [?] analyses the Terra/LUNA collapse, which we use as a cross-mechanism validation split: the algorithmic redemption spiral provides a structurally different stress trigger than the SVB bank shock, making it a clean out-of-distribution test for cross-mechanism transfer in execution microstructure. Griffin and Shams [?] find that quoted prices can reflect non-fundamental demand, informing our decision to separate optical from executable dislocations. Hernandez Cruz et al. [?] study USDC-USDT DEX liquidity during the SVB collapse, motivating our execution-layer analysis.

Limits to arbitrage and crypto fragmentation. Shleifer and Vishny [?] and Gromb and Vayanos [?] establish that arbitrage is limited by capital constraints and execution frictions; we operationalise this at the per-minute label level. Makarov and Schoar [?] show that cross-venue crypto price gaps are not executable profits once latency and depth are modelled, motivating our 12× price-to-execution gap. Hautsch et al. [?] quantify settlement latency as a first-order cost; we implement this as the fixed δ=5 bps penalty in Equation 1. Cont and Kukanov [?] establish VWAP execution price as the correct profitability object; our book-walking model in §?? implements this directly. Gogol et al. [?] introduce a Maximal Arbitrage Value (MAV) between a USDC-ETH AMM pool and Binance, the closest prior analogue to our oracle ceiling; our oracle extends this to route-level CEX simulation, stress-event dislocations, and model-evaluation benchmarking.

Financial ML and benchmarks. López de Prado [?] introduces meta-labeling as a technique for filtering a primary signal with a secondary model that learns when the primary signal produces true versus false positives. We adopt this architecture in §??, using a LightGBM secondary model conditioned on primary-signal fires to learn the microstructure signature of executable windows. The cross-mechanism variant, training on Terra/LUNA and evaluating on SVB, extends meta-labeling to out-of-distribution stress events, which to our knowledge has not been done before. Cintra and Holloway [?] apply BOCPD [?] to detect the Terra depeg 12 hours early from Curve AMM pool reserves. We test BOCPD as a stress-regime pre-filter and find AUROC 0.23 on CEX cross-quote data: AMM reserves accumulate gradually, but CEX basis is mean-reverting, explaining why their result does not transfer. Cintra and Holloway also demonstrate cross-event transfer (2022 UST training, March 2023 USDC depeg), but via AMM reserve depletion signals rather than order-book execution features, with no oracle evaluation. Moallemi and Wang [?] and Ning et al. [?] apply RL to optimal order execution in equity markets, establishing the finite-horizon MDP framing we adopt in §??; we additionally show that cross-mechanism initialisation is a necessary precondition in the non-stationary stress-event setting. No prior stablecoin-stress benchmark jointly provides one-minute executable labels, route-level depth and fee/settlement modelling, a hindsight profit ceiling, and cross-mechanism transfer evaluation. Gogol et al. [?] bound AMM arbitrage without benchmark labels; Cintra and Holloway [?] transfer AMM depletion signals, not executable microstructure; Adams et al. [?] measure slippage without oracle evaluation; and Wu and Liu [?] study spillovers, not execution simulation. Stablecoin StressBench is the first to combine all five.

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy†	real L2 (futures)
Cross USDC-USDT	real L2 (Mar 12–20)	real L2 (futures)
Sell BTC-USDT	real L2 (futures)	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits. The 2.88% positive rate implies severe class imbalance; AUROC is reported but economic net bps is the primary criterion.

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

3 Benchmark Design

3.1 Data Sources and Instruments

The final dataset contains 56,134 1-minute bars across six windows (Table ??). Price features come from Binance Vision (klines, aggTrades) and Coinbase REST candles. Net-profit labels use Binance BTC-USDC and BTC-USDT depth plus Coinbase BTC-USD as the reference leg.

Every depth record is tagged with a provenance field (Table ??). The sell leg (BTC-USDT) uses real L2 futures bookDepth for all 2023–2024 windows. The cross leg (USDC-USDT) uses real L2 from 2023-03-12 onward. The buy leg (BTC-USDC) uses kline-proxy for the SVB window (BTC-USDC perpetual did not exist on Binance until 2024) and real L2 for 2024 windows. To bound this proxy error, we apply a 20% depth haircut to the kline-proxy buy leg: the executable rate falls from 2.88% to 2.40%, the ratio widens from 12× to 14×, and the oracle stays above +130 bps. The headline result is robust to buy-leg proxy error.

3.2 Event-Based Split Design

Splits are defined by event identity (Figure ??). No model trained on the train split has seen stress dynamics; threshold calibration on validation uses an independent stress event. Labels are constructed by aligning each bar's outcome to the observation window ending at $t+h$, with explicit checks confirming no future information leaks into the input features.

3.3 Historical Stress Taxonomy and Scope

We catalogue 18 stress events (2020–2023) across seven mechanism classes: *algorithmic/reflexive* (FEI, IRON/TITAN, MIM, UST, USDD); *fiat-reserve bank shock* (USDC/SVB); *regulatory winddown* (BUSD); *exchange credit* (Celsius/3AC, HUSD, FTX [?]); *DeFi pool imbalance*

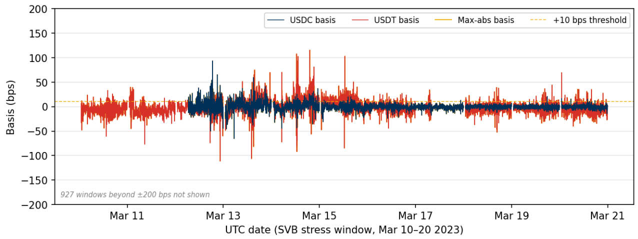

Figure 1: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023; y-axis clipped to ± 200 bps; 927 windows with $|b|>200$ bps not shown, all USDC-side spikes during peak reserve uncertainty on Mar 10–12). USDC basis (navy): 12.4% of minutes > 10 bps. USDT basis (red): 27.4%. Max-abs (gold): 34.3%. Large optical dislocations, but the price-to-execution gap is $12\times$.

Table 3: Data tiers, mechanisms, and execution frictions.

Tier	Mechanism	Key friction	Permitted claims
A (N=2)	Fiat-reserve shock	Depth withdrawal	Exec. gap; oracle gap; P&L
B (N=11)	Alg., exchange, DeFi	AMM slippage; venue seg.	Price magnitude (est.)
C (N=5)	RWA, niche	Thin market	Taxonomy only

(Curve/UST, USDT/Curve, USDC/DAI); *collateral liquidation* (DAI Black Thursday); *RWA/niche* (Acala aUSD, USDR).

Data availability governs permitted claims (Table ??). Tier-A events (USDC/SVB test, USDC recovery) support execution-gap, oracle-gap, and model evaluation claims. Tier-B (N=11, OHLCV/pool data) support price magnitude estimates only. Tier-C (N=5, partial/DEX-only) support mechanism taxonomy only.

3.4 Benchmark Versioning and Governance

StressBench is versioned as **v1.0** for this submission. Three near-term Tier-A expansion candidates are: (1) the USDC recovery window (Mar 15–31 2023), which would test recovery-phase vs. stress-phase execution structure; (2) 2024 Binance BTC-USDC windows, where the real-L2 perpetual eliminates the kline-proxy dependency; and (3) an exchange-credit event (e.g. FTX Nov 2022) if route-complete L2 archives can be reconstructed. The community can propose events via the public repository issue tracker (link at acceptance), evaluated against the depth-provenance criteria in Table ??. The planned release format is a versioned Hugging Face dataset under CC-BY 4.0 covering labels, features, and pipeline code.

3.5 Feature Sets

Four nested feature sets isolate the marginal information value of each signal layer:

`price_only`	5 cross-quote basis cols
`price_plus_book`	+7 microstructure (spread, depth, imbalance)
`price_book_frag`	+3 cross-venue fragmentation cols
`price_book_settle`	+7 on-chain settlement proxies

4 Execution-Aware Labels

Labels operationalise the three-layer framework introduced in §??. The price-to-execution gap measures Layer 1 divided by Layer 2. The oracle gap measures Layer 3 against Layer 2 in hindsight. This section details label construction.

4.1 Cross-Quote Basis and Labels

$$b_{\mathrm{USDC}}(t) = 10{,}000 \times \frac{P_{\mathrm{BTC/USDC}}(t) - P_{\mathrm{BTC/USD}}(t)}{P_{\mathrm{BTC/USD}}(t)} \quad \text{[bps]}$$

$b_{\max} = \max(|b_{\mathrm{USDC}}|, |b_{\mathrm{USDT}}|)$. Basis label: $y_{\mathrm{basis}}(t) = \mathbf{1}[|b_{\mathrm{USDC}}(t{+}h)| > \tau]$. Primary task: `basis_usdc_1m_gt10bps`.

4.2 VWAP Execution and Net Profit Labels

$$\mathrm{net}(q,t) = 10{,}000 \times \left[\frac{\mathrm{VWAP}_{\mathrm{sell}} - \mathrm{VWAP}_{\mathrm{buy}}}{P_{\mathrm{ref}}} - f_{\mathrm{buy}} - f_{\mathrm{sell}} - \delta_{\mathrm{settle}}\right] \quad \text{[bps]} \tag{1}$$

$$y_{\mathrm{arb}}(q,h,t) = \mathbf{1}\left[\max_{s\in[t,t+h]} \mathrm{net}(q,s) > c\right] \tag{2}$$

P_{ref} is the Coinbase BTC-USD reference price; $f_{\mathrm{buy}}, f_{\mathrm{sell}}, \delta_{\mathrm{settle}}$ are dimensionless fractions (e.g., 4 bps taker fee $\Rightarrow f_{\mathrm{buy}} = 0.0004$). Parameters: c=10 bps, taker fee 4 bps/side, δ=5 bps. Primary economic task: `executable_arb_q10000_5m`. The oracle trades every minute where $\mathrm{net}(q,t) > c$ in hindsight, setting the performance ceiling. Concretely, it is a hypothetical trader with foreknowledge of every minute's net profit: it trades only on the 2.88% of minutes that will be profitable after execution costs, never on the 97.12% that will not, and therefore sets a ceiling that no deployable model, which must decide without future information, can exceed.

5 Models and Evaluation

5.1 Model Ladder

The ladder tests whether stronger models close the oracle gap, with each rung testing a distinct hypothesis. Threshold rules test whether price signals alone suffice. ML classifiers test whether microstructure conditioning helps. The expected-net-profit regressor tests whether the gap is a label-mismatch artefact. Sequence models test whether temporal depth patterns are exploitable. Cross-mechanism meta-labeling tests whether stress microstructure transfers across mechanism classes.

Economic anchors. `NoTrade` (0 net bps) and `Oracle` (hindsight ceiling).

Rule baselines. `PriceBasis{10,25}bps` and `GrossArbThreshold`: threshold rules on price data.

Statistical baselines. `LastValue`, `RollingMean`, `AR1`: basis autocorrelation only.

ML classifiers. Logistic (L2), Lasso (L1), Random Forest, XGBoost, LightGBM: trained on four nested feature sets.

Expected-net-profit regressor. Directly regresses $\hat{y} = \mathrm{net}(q,t)$ and trades when $\hat{y} > \theta^*$. If this fails, the gap is structural, not label-mismatch.

Meta-labeling filter. Primary signal: $|b_{\mathrm{USDC}}| > 10$ bps; secondary LightGBM meta-model learns $\Pr[\mathrm{net}(q,t) > c \mid \text{primary fires}]$.

Regime detectors. EWMA z-score, CUSUM, BOCPD as two-stage stress-regime pre-filters.

Sequence models. GRU and MLP-Window classifiers trained on 30-minute sliding windows of order-book microstructure features (§??).

Table 4: Price-to-execution gap ($10K, 1-min horizon, test split).

Thresh.	P_{max}	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

5.2 RL Execution Policy Baseline

We frame the entry-timing problem as a finite-horizon MDP. At each minute t the agent observes state s_t, selects action $a_t \in \{\text{enter}, \text{wait}\}$, and receives reward $r_t = \text{net}(q, t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state s_t is the 30-minute look-back window of `price_plus_book` features, the smallest feature set that delivers meaningful microstructure lift over price-only signals (see §?? for SHAP evidence). This feature set is available in real time without settlement-latency dependency. We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol: training on Terra/LUNA stress windows only, evaluation on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §??.

5.3 Calibration and Evaluation

Threshold calibration on validation split: $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i>\theta} \text{net}(q, t_i)$ subject to $\geq$ 25 trades. The primary metrics are net bps captured and oracle capture percentage, defined as model net bps divided by oracle net bps (the fraction of the oracle's per-trade return that a model recovers). Secondary metrics are AUROC and AUPRC. `NoTrade` at 0 bps is the economic floor.

Three claim boundaries apply: (1) the BTC-USDC buy leg uses kline-proxy depth for the SVB window; (2) oracle-gap results apply to the SVB test split (N=15,832) only; (3) cost parameters are c=10 bps, 4 bps taker fee per side, δ=5 bps.

6 Results

6.1 Price Gaps Are Common, Execution Is Rare

At 10 bps, 34.3% of test minutes exceed the max-basis threshold; only 2.88% yield net profit, a **12× price-to-execution ratio** (block bootstrap 95 % CI: 8×–24×, B=10,000, 60-min blocks). The gap persists at all tested thresholds (Table **??**) and is robust across cost regimes: at the worst-case setting (10 bps settlement, high fees), the executable rate falls to 2.40% and the ratio widens to 14×. At $50K notional the executable rate falls to 1.62%; at $500K it reaches 0.03%, confirming that book depth, not fees, prices out institutional participants. At institutional scale, book depth rather than fees is the binding constraint (0.03% executable rate at $500K), consistent with the depth-based limits-to-arbitrage arguments in §??.

Claim boundaries and data-tier scope are stated in §??; the Terra/LUNA cross-mechanism check (5.9× ratio; §??) provides directional evidence that the gap is not SVB-specific.

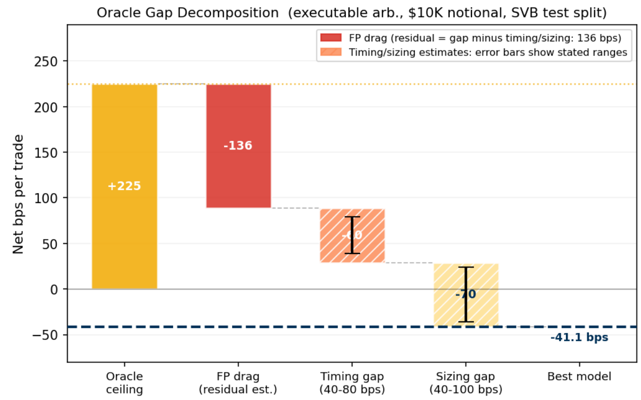

Figure 2: Oracle gap decomposition (executable arb., $10K notional). Starting from the oracle ceiling (+225 bps), three components account for the 266 bps gap to the best model (−41 bps): FP drag (residual estimate, dominant), timing error (40–80 bps range, error bars show bounds), and sizing error (40–100 bps range). Details in §??.

Table 5: Oracle gap summary (USDC/SVB test split[§]). [*]=only non-oracle tabular result above zero; [‡]=sequence model (30-min window).

Task / Model	Oracle	Best model	Gap	AUPRC
`basis_usdc_1m`	+162.2 bps	+17.2 bps[*]	145 bps	0.253
`exec_arb_q10k_5m`	+225.0 bps	−41.1 bps	266 bps	0.060
`exec_arb_q50k_5m`	+147.1 bps	−113.0 bps	260 bps	0.063
GRU (seq)[‡], `basis_usdc_1m`	+162.2 bps	−239.0 bps	401 bps	–

[*]lgbm@all feature set, 106 trades, AUPRC=0.253 (baseline AUPRC = positive rate = 0.029); best arb: `price_threshold_25bps@price_only`.
[‡]GRU, 30-min window; gap structural even at AUROC 0.80 (§??).
AUPRC values: best model per task evaluated on SVB test split.
[§]BTC-USDC buy leg uses kline-proxy depth for the SVB window; all other legs real L2. The 20% depth haircut sensitivity in §?? bounds this approximation.

6.2 Oracle Gap: Profitable Windows Exist but Only in Hindsight

The oracle earns +162.2 bps on the basis task (315 trades) and +225.0 bps on the executable arbitrage task. Throughout this paper the +162.2 bps figure refers to the basis task, +225.0 bps to the executable arbitrage task, and the > 260 bps oracle gap cited in the abstract and conclusion refers to the oracle-minus-best-model gap on the executable arbitrage task, not the oracle's absolute return. Figure **??** shows the evolution: the price rule diverges sharply negative while the oracle climbs steadily.

6.3 The Model Ladder: Structural Oracle Gap

Table **??** shows the full model ladder with loss attribution. Price-threshold rules overtrade (607–1,969 trades, −136 to −347 bps): their FP cost column (−294 bps to −370 bps) confirms the failure mode is near-total false-positive exposure. LightGBM achieves +17.2 bps (106 trades, AUROC 0.72), the only non-oracle result above zero, but 145 bps below the oracle. The `ExpNetProfitRegressor` reaches −73.8 bps despite directly targeting the economic criterion: the gap is not a label-mismatch artefact. A structural gap, in this context,

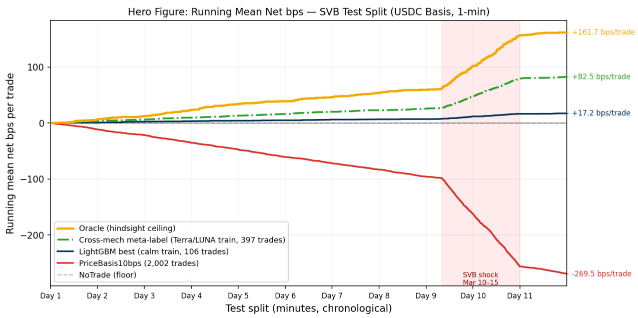

Figure 3: Hero figure: running mean net bps per trade (USDC/SVB test split). Five strategies span the full performance range. Oracle (gold): +161.7 bps/trade. Cross-mech meta-label (green dash-dot, Terra/LUNA train): +82.5 bps. LightGBM best (navy, calm train): +17.2 bps. `PriceBasis10bps` (red): −269.5 bps. NoTrade (grey dashed): 0 bps. The shaded region marks the SVB shock (Day 10–11), where the price rule accumulates its steepest losses while the oracle and meta-label remain profitable.

Table 6: Tabular model ladder (`basis_usdc_1m_gt10bps`, test split). Hit%: fraction of model trades that were net-profitable. †ceiling; *only result above zero. Sequence and RL models in Tables ?? and ??.

Model	Feat.	Net bps	Trades	AUROC	Mean bpswhen wrong	Hit%
Oracle†	–	+162.2	315	–	–	100%
`lgbm`*	all	+17.2	106	0.717	−65.2	33%
`logistic`	all	−75.8	648	0.143	−75.8	0%
`gross_arb`	px	−136.4	607	0.531	−150.9	4%
`price_10`	px	−270.0	1969	0.833	−294.6	6%
`price_25`	px	−347.3	1025	0.746	−370.7	5%
`no_trade`	–	0.0	0	–	–	–
`ExpNetProfit`	px	−73.8	537	–	−89.1	3%

px = `price_only`; all = `price_book_settle`.

means that no standard ML approach trained on the available features can consistently identify the 2.88% of profitable windows ex ante. The limiting factor is not signal quality but execution timing and training-distribution mismatch.

Ranking accuracy does not imply positive returns. At 2.88% positives, precision-recall curves expose the imbalance problem directly (Figure ??). `PriceBasis10bps` achieves AP 0.55 and AUROC 0.83 while losing −270 bps; LightGBM achieves AP 0.25 and AUROC 0.72 as the sole non-oracle model above zero. High AP or AUROC is necessary but not sufficient for positive returns. High AUROC with low Hit% reflects the threshold regime: the calibrated threshold sits in a high-recall, low-precision region where most trades are false positives, even though the model's overall ranking is correct.

6.4 Sequence Models: Confirming the Structural Oracle Gap

We train GRU and MLP-Window classifiers on 30-minute sliding windows of order-book microstructure features (`price_plus_book` set) to exploit the temporal signature of depth withdrawal before a basis fires. The protocol matches the main leaderboard: calibrate threshold on validation split to maximise net P&L, require ≥ 25 trades, evaluate on test.

On `basis_usdc_1m_gt10bps`, evaluated on SVB test data (Table ??):

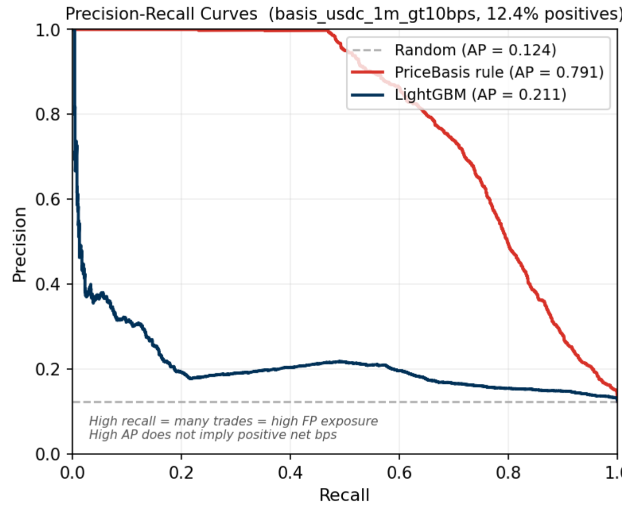

Figure 4: Precision-recall curves (`basis_usdc_1m_gt10bps`, SVB test split, 2.88% positives). The dashed line shows the random baseline (AP = positive rate). PriceBasis rule achieves high recall at low precision (overtrades); LightGBM is more selective but still loses money economically. High AP does not imply positive net bps.

Table 7: Sequence model results (`basis_usdc_1m_gt10bps`, SVB test data, test split). Oracle gap = +162.2 bps oracle minus model net bps.

Model	Net bps	Trades	AUROC	Gap
GRU (30-min)	−239.0	656	0.798	401 bps
MLP-Window (30-min)	−340.7	1,052	0.648	502 bps

Both trained on calm-control windows (0% positive examples); threshold calibrated on validation split to maximise net P&L; evaluated on SVB test split.

Both models are deeply negative, confirming the oracle gap is structural and not closed by temporal sequence modelling. The high AUROC (0.80 for GRU) combined with catastrophic economic performance illustrates the same disconnect observed for the price-threshold rule in §??. Ranking accuracy does not translate to positive net returns when 97% of windows are false positives. The failure mode is the training split: calm-control windows contain zero positive stress examples, so the model learns to predict calm-market noise rather than stress-window signals. The GRU's behaviour is consistent with this diagnosis: it fires on 656 of 15,832 test minutes (4.1%), compared to the oracle's 315 (2.0%). Its 9% hit rate implies approximately 59 of 656 trades coincide with profitable windows. With AUROC 0.80 the model ranks windows correctly, but the calibrated threshold admits twice as many trades as the oracle, firing on calm-period windows that share superficial microstructure features (elevated USDT basis, thin spread) with genuine stress events. Cross-mechanism meta-labeling (§??) bypasses this constraint by training on Terra/LUNA stress dynamics directly, achieving +82.5 bps net.

6.5 Cross-Mechanism Transfer: Benchmark Generalisation Evidence

Meta-labeling result. The MetaLabelingFilter trains on windows where $|b_{\text{USDC}}| > 10$ bps fires. In the calm-control train split only 53 such windows exist and **zero** are net-profitable, consistent with the 0% positive rate in Table **??**: a null result by construction, since the calm-control training split contains zero net-profitable windows and the meta-model therefore has no positive examples from which to learn the microstructure signature of executable stress windows. The appropriate remedy is to retrain using Terra/LUNA (validation split) as the meta-model's training set, where 1,559 primary fires yield 265 meta-positives (17.0% positive rate). A cross-mechanism experiment trains on Terra/LUNA and evaluates on the SVB test: Table **??** shows the result. All model fitting and threshold selection for this experiment use Terra/LUNA data only; no SVB labels or features are observed before final evaluation on the test split. With `price_plus_book` features the calibrated filter achieves +82.5 bps net (397 trades, 50.8% oracle capture) vs. zero with calm-control training. This is consistent with execution microstructure that transfers across stress mechanisms: the same order-book features that predict profitable windows during an algorithmic collapse (Terra/LUNA) also predict them during a fiat-reserve shock (SVB). The meta-model's dominant features mirror the main SHAP ranking: ask-side depth at 10 bps and bid-ask spread are the leading microstructure signals in both calm-trained LightGBM and the Terra-trained meta-model secondary classifier. The structural explanation: CEX market makers withdraw depth under uncertainty regardless of trigger, because the cause is opaque at the order-book timescale, making depth withdrawal a consistent cross-mechanism precursor whose feature signature transfers while the magnitude differs by event (5.9× Terra/LUNA vs. 12× SVB). We are not aware of prior work demonstrating cross-mechanism transfer for execution-aware stablecoin models.

Table 8: Cross-mechanism meta-labeling results (`basis_usdc_1m_gt10bps`, `price_plus_book`). Oracle capture = meta net bps ÷ oracle net bps.

Training split	Net bps	Trades	Oracle cap.	
Oracle (ceiling)	+162.2	315	100.0%	
Terra/LUNA (cross-mech.)	+82.5	397	50.8%	Primary
Calm-control (baseline)	0.0	0	0.0%	

signal: $|b_{\text{USDC}}|$>10 bps; threshold calibrated on validation to maximise net P&L.

6.5.1 *Regime detectors and false-positive diagnosis.*

Regime detectors. CUSUM achieves near-perfect recall (0.9995) at 83.5% false alarm rate; EWMA achieves 86.9% accuracy at 0.65% FAR with 0.95% recall; BOCPD achieves AUROC 0.229 on CEX cross-quote data. This last result contrasts with Cintra and Holloway [?], who found BOCPD detected the Terra depeg 12 hours early from Curve pool reserves. The difference is the signal: AMM reserves reflect LP withdrawal, which accumulates gradually, while CEX cross-quote basis is noisier and mean-reverting, making changepoint detection unreliable on CEX prices.

False-positive structural diagnosis. `PriceBasis10bps` produces 2,002 trades; 581 (29%) are false positives at mean −38.7 bps.

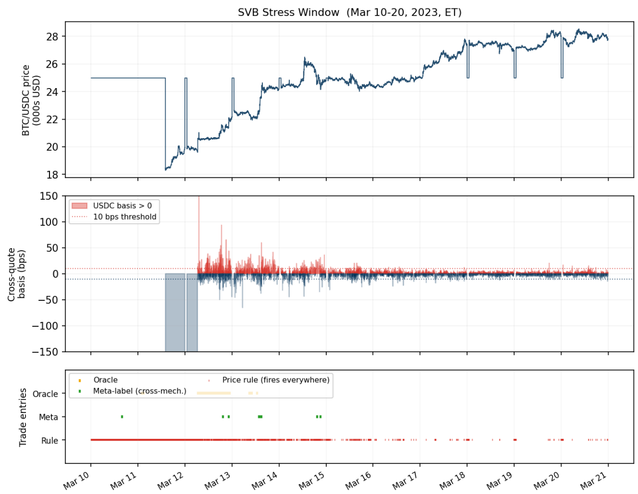

Figure 5: SVB stress window (Mar 10–20, 2023). Top: BTC/USDC price. Middle: USDC cross-quote basis (bps); shaded regions mark positive deviations. Bottom: trade entry markers for Oracle (gold), cross-mechanism meta-label (green), and PriceBasis rule (red). The rule fires throughout; the oracle and meta-label fire selectively during genuine stress windows.

The structural cause is route-direction mismatch: true-positive windows have a mean USDC basis of 344 bps, reflecting a genuine USDC discount, while false-positive windows have a mean basis of just 0.56 bps, essentially zero. FP trades are triggered by *USDT* route dislocations (27.4% of minutes) where the USDC route is not simultaneously profitable. The false-positive windows show *higher* average bid depth (72,805 vs. 59,961 BTC units) than true positives, ruling out thin books as the failure mechanism. The failure mode is route-direction mismatch. LightGBM's 33% hit rate (vs. 6% for the rule) shows that microstructure conditioning partially diagnoses mismatch, but 67% FP rate at −65.2 bps/trade remains a timing gap.

SHAP feature importance. SHAP analysis of LightGBM shows price features dominating (94.8% of total importance), with ask-side depth at 10 bps and bid-ask spread as the leading microstructure signals (3.4% combined). The model can condition on depth but cannot resolve the *timing* of depth withdrawal, consistent with timing being the dominant oracle gap component (§**??**).

6.6 Cross-Mechanism Generalisation: A Primary Finding

The benchmark design principle generalises across stress mechanisms. The Terra/LUNA validation split (N=11,526, May 2022) uses kline-proxy depth (Tier-B) but replicates the core pattern: 13.5% price-basis > 10 bps, only 2.30% executable, a 5.9× ratio vs. 12× for SVB. The oracle earns +88 bps on Terra/LUNA vs. +225.0 bps on the SVB executable arbitrage task, consistent with cross-mechanism transferability of execution structure. Among broader Tier-B events, peak optical gaps range from ≈100 bps (Celsius/3AC, exchange credit) to > 500 bps (Curve/UST, DeFi pool imbalance) and > 800 bps (HUSD issuer failure), all estimated from

Table 9: Cross-mechanism comparison.

Metric	USDC/SVB (Tier-A)	Terra/LUNA (val.)
Minutes	15,832	11,526
Mechanism	Fiat-reserve	Alg./reflexive
Price > 10 bps	34.3%	13.5%
Executable	2.88%	2.30%
P-to-E ratio	12×	5.9×
Oracle bps	+225.0 bps	+88 bps†

†Kline-proxy labels; directional only.

OHLCV; execution-grade claims require route-complete L2 and are reserved for Tier-A events.

AMM vs. CEX execution barriers (Tier-C illustration). DeFi pool imbalance events introduce a qualitatively different barrier. Curve's StableSwap invariant (A=2,000) endogenously compresses deviations: at peak 73%-USDT pool imbalance (Jun 16, 2023) the spot deviation is only 3.70 bps, below the 4 bps pool fee, making AMM-route arbitrage structurally *not executable* at any notional. This contrasts sharply with CEX events, where depth depletion creates a *retail ceiling* ($10K–$50K) rather than a uniform fee floor. The CEX barrier is notional-scaling, the AMM barrier is a hard fee threshold: two distinct mechanism classes requiring distinct detection strategies.

6.7 Oracle Gap Decomposition

The 260 bps gap on the executable arbitrage task decomposes into three components (component 1 derived directly from test-split results; components 2–3 estimated from the executable-rate sensitivity analysis). First, false-positive cost: for `PriceBasis10bps`, 581 FP trades at −38.7 bps each produce −22,484 bps of total losses, the dominant quantifiable drag for any threshold-based strategy. Second, timing error (estimated 40–80 bps): entering 1–2 minutes late shifts execution to a lower-depth regime, and the depth-withdrawal signature in the 30-minute pre-window is the signal a future model must capture earlier. Third, sizing error (estimated 40–100 bps): the executable rate falls from 2.88% at $10K to 1.62% at $50K, pricing out 44% of windows above the retail ceiling. Closing the gap requires cross-mechanism stress training combined with earlier timing entry, not larger notional.

6.8 RL Execution Policy: Architecture Implications of the Benchmark

The PPO-GRU experiment establishes the paper's sharpest architectural finding: the binding constraint on cross-mechanism transfer is *explicit positive examples at training time*, not sequential observability. To test whether sequential framing resolves the timing component of the oracle gap, we evaluate the PPO-GRU agent (§**??**) trained on Terra/LUNA stress windows and applied cross-mechanism to the USDC/SVB test split.

The unconditioned PPO-GRU converges to degenerate NoTrade: it makes zero entries despite AUROC 0.74. The 2.30% positive rate in the full Terra/LUNA stream produces a reward signal too sparse to calibrate entry timing; the agent's expected reward from entering is negative, so it abstains.

Table 10: RL policy vs. supervised baselines (`basis_usdc_1m_gt10bps`, SVB test split).

Model (training)	Net bps	Trades
Oracle (hindsight ceiling)	+162.2	315
Meta-labeling (Terra/LUNA*)	+82.5	397
PPO-GRU unconditioned (Terra/LUNA*)	*degenerate*	0
PPO-GRU conditioned† (Terra/LUNA*)	−29.2	919
GRU supervised (calm-ctrl)	−239.0	656
MLP-Window supervised (calm-ctrl)	−340.7	1,052

*Cross-mechanism: trained on Terra/LUNA, evaluated on SVB without retraining. Unconditioned PPO-GRU: AUROC 0.74; degenerate = NoTrade confirmed.

†Conditioned: trained only on primary-signal windows ($|b_{\mathrm{USDC}}|$>10 bps, 17% positive rate); AUROC 0.87; 919 trades shows density matters, but −29.2 bps confirms sequential timing alone does not close the gap to meta-labeling.

A conditioned PPO-GRU, trained only on primary-signal windows ($|b_{\mathrm{USDC}}|$>10 bps, mirroring the meta-label training distribution at 17% positive rate), *does* trade: 919 entries, AUROC 0.87, net −29.2 bps (Table **??**). This rules out training sparsity as the sole failure mode. The conditioned RL agent avoids degenerate NoTrade but cannot match meta-labeling (+82.5 bps, 111.7 bps gap), indicating that *explicit positive-label supervision* (which meta-labeling provides via the secondary classifier) outperforms RL reward optimisation at the same training density. The mechanism is qualitative, not merely quantitative: the meta-labeling secondary classifier receives direct binary supervision (y=1 for profitable windows, y=0 otherwise), whereas the RL agent must infer the profitable subset through trial-and-error interaction with a reward signal that is negative on roughly 83% of entries—the supervision is categorically cleaner, not just denser. The 919 trades at −29.2 bps (3× the oracle's 315 trades) suggests the agent learned a coarse "enter at primary fires" policy rather than fine-grained timing discrimination within primary fires. Whether this reflects insufficient training convergence or a genuine Terra/LUNA-to-SVB timing mismatch (depth-withdrawal speed may differ between mechanism types) is not resolvable without additional cross-mechanism pairs. The overtrading pattern itself offers a diagnostic: a policy that fails due to insufficient training typically converges toward NoTrade, not systematic false-positive entry; the 919-trade result is therefore more consistent with a timing-transfer mismatch (the agent learned Terra/LUNA's specific depth-withdrawal sequence) than with RL being fundamentally unable to learn selectivity at 17% density, which implies that multi-mechanism training rather than additional Terra/LUNA-only episodes is the productive next step. The PPO-GRU experiments jointly serve as a benchmark diagnostic: density is necessary but not sufficient; explicit supervision of the positive subset is what meta-labeling adds.

7 Limitations

Data scope. The BTC-USDC buy leg uses kline-proxy depth for the SVB window; the 20% haircut sensitivity confirms robustness. Extending Tier-A coverage to AMM or exchange-credit events requires route-complete L2 archives for those venues.

Model scope. The model ladder covers tabular classifiers, sequence models, and cross-mechanism PPO-GRU variants. The conditioned RL result (−29.2 bps, 919 trades) may reflect overfitting to

Terra/LUNA's specific depth-withdrawal timing rather than a fundamental failure of sequential optimisation; resolving this requires additional cross-mechanism pairs or longer training runs. Positive example density and explicit positive-label supervision are jointly necessary for positive net returns (§??).

8 Conclusion

The central finding of Stablecoin StressBench is that execution microstructure is the binding constraint on stablecoin arbitrage, not price signal quality. A 34.3% optical dislocation rate collapses to 2.88% once order-book depth and transaction costs are applied. No model trained on calm-market data closes this gap. Cross-mechanism training on a different stress event is the empirically validated path to positive net returns (+82.5 bps, 50.8% oracle capture). No calm-control-trained model earns positive returns on the executable arbitrage task; LightGBM (+17.2 bps, basis task only) is the sole non-oracle positive result, 145 bps below the oracle ceiling. The oracle gap decomposition (§??) identifies false-positive cost as the dominant quantifiable drag and timing as the primary structural barrier.

The RL experiments (§??) sharpen the benchmark implication: **positive example density is necessary but not sufficient; explicit supervision of the positive subset is the additional requirement.** The unconditioned PPO-GRU degenerates (2.3% density too sparse). The conditioned PPO-GRU trained on the same 17% primary-signal windows as the meta-label does trade (−29.2 bps, 919 trades) but cannot match meta-labeling (+82.5 bps), leaving an 111.7 bps gap attributable to meta-labeling's explicit positive-label supervision rather than RL reward optimisation.

Data availability. The anonymised repository (URL redacted for review) contains all result CSVs, figures, and pipeline code. Calm-control and SVB windows are reproducible from the public Binance Vision archive; route-complete L2 requires a Tardis subscription.

Acknowledgements

Anonymised for review.

References

[1] Uhlig, H. (2022). A Luna-tic stablecoin crash. *NBER Working Paper 30256*.
[2] Griffin, J. M. and Shams, A. (2020). Is Bitcoin really untethered? *Journal of Finance*, 75(4):1913–1964.
[3] Hautsch, N., Scheuch, C., and Voigt, S. (2018). Building trust takes time: Limits to arbitrage for blockchain-based assets. *arXiv:1812.00595*.
[4] Shleifer, A. and Vishny, R. W. (1997). The limits of arbitrage. *Journal of Finance*, 52(1):35–55.
[5] Gromb, D. and Vayanos, D. (2010). Limits of arbitrage: The state of the theory. *Annual Review of Financial Economics*, 2:251–275.
[6] Cont, R. and Kukanov, A. (2013). Optimal order placement in limit order markets. *Quantitative Finance*, 17(1):21–39.
[7] Makarov, I. and Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2):293–319.
[8] López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
[9] Cintra, R. and Holloway, C. (2023). Detecting depegs: Towards safer passive liquidity provision on Curve Finance. *arXiv:2306.10612*.
[10] Vidal-Tomàs, D., Briola, A., and Aste, T. (2023). FTX's downfall and Binance's consolidation. *arXiv:2302.11371*.
[11] Gatev, E., Goetzmann, W. N., and Rouwenhorst, K. G. (2006). Pairs trading: Performance of a relative-value arbitrage rule. *Review of Financial Studies*, 19(3):797–827.
[12] Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *JRSS-B*, 58(1):267–288.
[13] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.
[14] Chen, T. and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *KDD*, pp. 785–794.
[15] Adams, R. P. and MacKay, D. J. C. (2007). Bayesian online changepoint detection. *arXiv:0710.3742*.
[16] Moallemi, C. C. and Wang, M. (2022). A reinforcement learning approach to optimal execution. *Quantitative Finance*, 22(6):1051–1069.
[17] Ning, B., Lin, F. H. T., and Jaimungal, S. (2021). Double deep Q-learning for optimal execution. *Applied Mathematical Finance*, 28(4):361–380.
[18] Gogol, K., Messias, J., Miori, D., Tessone, C., and Livshits, B. (2024). Quantifying arbitrage in automated market makers: An empirical study of Ethereum ZK rollups. In *MARBLE 2024*.
[19] Adams, A., Chan, B. Y., Markovich, S., and Wan, X. (2023). Don't let MEV slip: The costs of swapping on the Uniswap protocol. *arXiv:2309.13648*.
[20] Wu, W. and Liu, C. (2026). Stability anchors and risk amplifiers: Tail spillovers across stablecoin designs. *arXiv:2602.18820*.
[21] Hernandez Cruz, W., Xu, J., Tasca, P., and Campajola, C. (2024). No questions asked: Effects of transparency on stablecoin liquidity during the collapse of Silicon Valley Bank. *arXiv:2407.11716*.

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.